
FedReCall: Recalling Client-Specific Directions in Federated LoRA Fine-tuning

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Abstract

1 Low-rank adaptation (LoRA) enables communication-efficient federated fine-
2 tuning of large language models, but standard FedLoRA averages the factors
3 A and B separately even though the model update is determined by their product
4 $\Delta W = sBA$. Under task heterogeneity, factor-wise aggregation can further dilute
5 directions that are locally useful but not strongly shared across clients; we call
6 this phenomenon *directional dilution*. We analyze this effect through the singular
7 directions of client LoRA updates and find that low-strength directions are most
8 vulnerable to dilution. Their repeated owner-specific recovery suggests client-
9 specific adaptation signals rather than random noise. We propose FEDRECALL, a
10 client-private plug-in that captures observable dilution gaps, selects reliable diluted
11 directions into a private frozen cache, and recalls them through a low-rank bypass
12 guarded by a single-batch loss probe when the client next participates. As a plug-
13 in, FEDRECALL can be seamlessly integrated into existing FedLoRA methods
14 without changing their aggregation rules, communication payloads, or trainable-
15 parameter sets, consistently improving personalized adaptation and improving
16 global evaluation metrics on most aggregators.

17 1 Introduction

18 Large language models (LLMs) are increasingly adapted to specialized tasks, but full fine-tuning
19 remains expensive at scale [32, 9]. Low-rank adaptation (LoRA) [12] offers a practical alternative by
20 training a compact update $\Delta W = \frac{\alpha}{r}BA$ while keeping the backbone frozen. When task data is dis-
21 tributed across privacy-sensitive clients, federated LoRA (FedLoRA) further reduces communication
22 because clients exchange only lightweight adapter factors rather than full model weights [23, 8].

23 Existing FedLoRA methods mainly improve how LoRA factors are aggregated or decide which
24 components remain private [39, 30, 3, 34, 2, 15, 28]. Personalized FL offers another route, but
25 typical PFL designs introduce extra local modules, optimization stages, or personalization-specific
26 structures [1, 31, 17, 27]. These approaches improve different parts of the overhead–capability
27 trade-off, as summarized in Table 1, but they do not directly ask which directions aggregation actually
28 erases.

29 Federated learning compresses multiple client objectives into a single shared update, so task
30 heterogeneity naturally creates *update conflict*: directions with broad cross-client support tend
31 to dominate, while directions that are useful to one client but weakly shared are more likely
32 to be suppressed. In FedLoRA, this conflict is amplified by factor-wise aggregation. The
33 server averages A and B separately, but model behavior is governed by the effective update
34 $\Delta W = sBA$; therefore, aggregation can retain different local directions to very different de-
35 grees. We call the resulting selective attenuation of locally useful directions *directional dilution*.

Our key insight is that the directions most vulnerable to aggregation are not arbitrary noise, but recurring client-specific adaptation signals. As shown in Figure 2, these directions suffer larger relative dilution, are repeatedly recovered by local training and suppressed again by aggregation, and correspond to the fragile client-specific components that standard sharing tends to smooth, distort, or cancel (Figure 2c). Existing methods do not explicitly identify and locally restore this diluted directional signal.

We propose *FedReCall* (*Federated Re-injection of Client-Adaptive Lost directions*) to answer this question with a client-private recall mechanism. After each aggregation, a participating client decomposes its effective LoRA update, measures how much the new global update retains along each local direction, and converts the missing positive strength into an observable dilution gap. Reliable diluted directions are stored as a private frozen cache and recalled as a low-rank bypass when the client next participates; a single-batch quality gate enables the bypass only when it improves the client’s local loss. The cache is never uploaded, so clients still communicate only clean LoRA factors and the server runs the original aggregation rule unchanged. FedReCall therefore adds personalized memory without extra trainable parameters, communication payload, or a new server-side optimizer.

Mechanism diagnostics and main experiments confirm that this local recall reduces directional dilution, dampens the recovery–disruption cycle, and improves both personalized client quality and global evaluation metrics across representative FedLoRA aggregators (Figure 1). As Table 1 highlights, it does so as a plug-in without changing the aggregation rule, communication payload, or trainable-parameter set. Together, these results show that diluted client-specific directions can be recovered without extra communication or trainable parameters, yielding stronger personalized and global performance while preserving the original federated training protocol.

Contributions.

- We uncover directional dilution, a structured form of personalization loss where factor-wise aggregation repeatedly weakens client-specific directions in the effective LoRA update space.
- We propose FedReCall, a client-private recall plug-in that restores diluted directions with an observable dilution gap, a frozen low-rank bypass, and a lightweight quality gate.
- We show that FedReCall improves personalized and global performance across representative FedLoRA aggregators while preserving the original server rule, communication payload, and trainable-parameter set.

2 Related Work

Parameter-Efficient Fine-Tuning Parameter-efficient fine-tuning (PEFT) aims to adapt a large pre-trained model to downstream tasks while updating only a small subset of task-specific parameters and keeping the backbone largely frozen[9]. A variety of PEFT techniques have been explored, including prompt-based tuning[18, 16, 14], ladapter-based methods[11], low-rank adaptation methods such as LoRA[12, 22, 24], and so on. In this paper, we focus on LoRA, as it has become one of the most widely used PEFT methods for large models due to its strong adaptation performance, parameter efficiency, and broad applicability across downstream tasks[12, 8, 7, 21]. While we adopt LoRA’s structural assumptions, our focus is not on LoRA in isolation but on a federated-specific failure

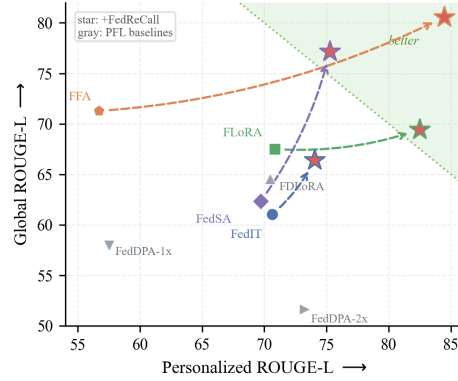


Figure 1: Performance balance between client-side and global performance.

Table 1: Prerequisites and application scope of FedLoRA methods.

Method	Extra-param free	Extra-optim free	Extra-comm free	Personalization	Plug-in
FedIT	✓	✓	✓	✗	–
FFA-LoRA	✓	✓	✓	✗	✗
FedSA-LoRA	✓	✓	✓	✗	✗
FLoRA	✓	✓	✗	✗	✗
FDLoRA	✗	✗	✓	✓	✗
FedDPA-T	✗	✗	✓	✓	✗
FedReCall	✓	✓	✓	✓	✓

mode that emerges only when the bilinear LoRA product BA is split across clients and aggregated factor-by-factor.

Parameter-Efficient Fine-Tuning in Federated Learning Recent work has introduced PEFT into federated learning to make the fine-tuning of pre-trained or foundation models feasible under limited client-side computation, memory, and communication budgets, since only a small subset of parameters needs to be optimized and exchanged[29, 3]. Existing studies have explored several PEFT mechanisms in federated settings, including prompt-based tuning[20], adapter-style modules[37, 35], and LoRA-based fine-tuning[3, 40, 34, 2]. Among them, LoRA has become a particularly active direction in federated fine-tuning, with prior work addressing model heterogeneity and personalization[40], aggregation accuracy for heterogeneous LoRA adapters[4], privacy-preserving and more stable LoRA optimization[30, 13], and robustness under increasing data heterogeneity[15, 3].

Personalized Federated Learning Personalized federated learning (PFL) studies how to tailor federated models to individual clients under statistical heterogeneity. Because client data distributions often differ substantially, a single global model may not perform equally well for every client[1, 6, 31, 17]. Existing PFL methods address this issue in several ways. Some split the model into globally shared and client-specific components, such as keeping a shared backbone with local personalization layers[1, 25] or learning a shared representation with client-specific heads[5, 36]. Others learn personalized models together with shared knowledge, for example by adaptively combining local and global models[6, 10], or by learning personalized weighted combinations of available models[43], decoupling personalized optimization from global model learning in a bi-level formulation[31], or improving fairness and robustness through personalization[17, 19].

3 Method

3.1 Preliminaries and notations

Consider a federated learning system with N clients, each holding a private dataset $\mathcal{D}_c$. All clients share a pre-trained backbone W_{base} and fine-tune it via LoRA [12]: each LoRA-adapted layer l adds a low-rank update $\Delta W_l = s_l B_l A_l$, where $B_l \in \mathbb{R}^{d_{\text{out}} \times r}$, $A_l \in \mathbb{R}^{r \times d_{\text{in}}}$, and $s_l = \alpha_l/r$ is the LoRA scaling factor. In each communication round t , a subset $S_t \subseteq [N]$ of clients is selected. Each selected client c downloads the current global LoRA factors, performs local training, and uploads the trained factors $(A_{c,l}^t, B_{c,l}^t)$.

The server aggregates LoRA factors using the standard factor-wise FedIT protocol:

$$A_l^{t,\text{glob}} = \sum_{j \in S_t} p_j A_{j,l}^t, \quad B_l^{t,\text{glob}} = \sum_{j \in S_t} p_j B_{j,l}^t, \quad \sum_{j \in S_t} p_j = 1. \quad (1)$$

This produces the actual global LoRA update used in the next round,

$$\Delta W_l^{t,\text{glob}} = s_l B_l^{t,\text{glob}} A_l^{t,\text{glob}}. \quad (2)$$

Because the LoRA product is bilinear, factor-wise averaging does not preserve the effective update:

$$s_l B_l^{t,\text{glob}} A_l^{t,\text{glob}} \neq \sum_{j \in S_t} p_j s_l B_{j,l}^t A_{j,l}^t \quad (3)$$

in general, so individual client directions can be selectively lost in the aggregated update even when both factors are faithfully averaged.

Direction decomposition. Each client’s effective LoRA update admits a singular value decomposition

$$\Delta W_{c,l}^t = \sum_{i=0}^{r-1} \sigma_i u_i v_i^\top, \quad (4)$$

where $\sigma_0 \geq \dots \geq \sigma_{r-1} \geq 0$ are the singular values (amplitudes) and $u_i \in \mathbb{R}^{d_{\text{out}}}$, $v_i \in \mathbb{R}^{d_{\text{in}}}$ are the corresponding singular vectors. We call each rank-one atom $\phi_i = u_i v_i^\top$ a *direction* in the update space; because u_i and v_i are columns of orthonormal bases, $\|\phi_i\|_F = 1$ and distinct atoms are Frobenius-orthogonal. The Frobenius inner product $\langle M, \phi_i \rangle_F = u_i^\top M v_i$ measures how much of direction ϕ_i is present in any matrix M . We group directions by index position $q_i = i/r$ into four bands: *head* ($q < 0.2$), *middle* ($0.2 \leq q < 0.6$), *tail* ($0.6 \leq q < 0.9$), and *floor* ($q \geq 0.9$). The head band carries the largest-amplitude directions; the floor band carries the weakest non-trivial ones.

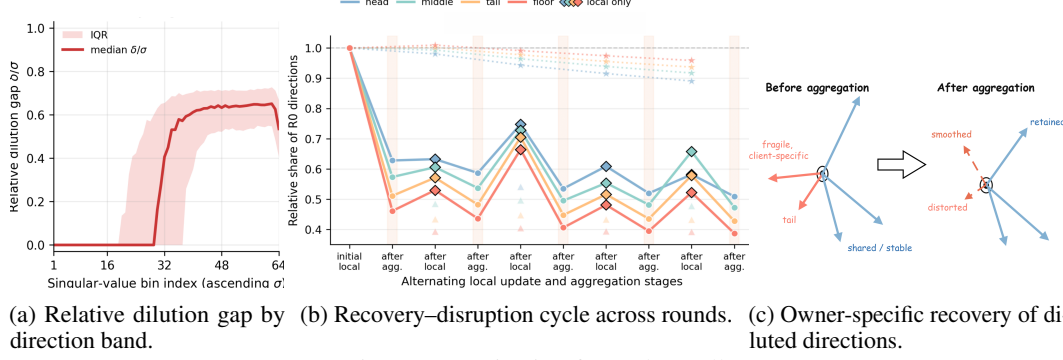


Figure 2: Motivation for FedReCall.

3.2 Motivation: directional dilution under factor-wise aggregation

Each direction ϕ_i in a client’s update carries a portion of its adaptation signal, weighted by its amplitude σ_i . Under factor-wise averaging (Eq. 3), the resulting global update does not preserve these client directions equally: large-amplitude (head) directions are largely shared across clients and survive aggregation; small-amplitude (tail/floor) directions, by contrast, encode client-specific signal; their lower amplitude leaves them more vulnerable to bilinear distortion—an asymmetry also documented in the model-merging literature [41].

To measure this vulnerability precisely, we define two per-direction quantities. The *global retention* of direction ϕ_i is

$$\alpha_i = \langle \Delta W_l^{t,\text{glob}}, \phi_i \rangle_F = s_l u_i^\top B_l^{t,\text{glob}} (A_l^{t,\text{glob}} v_i), \quad (5)$$

and the *dilution gap* is the local strength missing from this projection,

$$\delta_i = \max(0, \sigma_i - \alpha_i^+), \quad \alpha_i^+ = \max(\alpha_i, 0). \quad (6)$$

The relative gap $\delta_i/\sigma_i \in [0, 1]$ serves as a scale-invariant diagnostic; the absolute gap δ_i will be reused as the cache amplitude in Section 3.3. We make two observations: *which* directions are most affected (Obs. 1), and *what role* they play (Obs. 2).

Observation 1: Dilution is structured by within-client direction rank. Grouping directions by their rank position inside each client’s update, the median δ_i/σ_i rises monotonically from head to floor (Figure 2): the lower a direction sits in its owner’s ranking, the larger the share of its strength removed by factor-wise averaging.

Observation 2: The diluted directions encode client-specific adaptation knowledge. Three independent lines of evidence show that the directions most affected by aggregation carry functionally meaningful, client-owned signal. (i) *Persistence under local training.* When aggregation is turned off (dashed lines in Figure 2), the band ordering *reverses*: tail and floor directions are preserved best, while head and middle decay fastest. Since SGD on a client’s own data retains whatever actively reduces the local loss, the persistence of tail/floor directions confirms that they contribute to the client’s task performance. (ii) *Owner specificity.* Triangles in Figure 2 show that a direction defined by client c is recovered better by c in subsequent rounds than by any other client, with the specificity gap largest in tail/floor. These directions therefore encode adaptation signal that is client-specific rather than population-shared. (iii) *Loss-blind validation.* Section 4.3 provides an out-of-sample test that further isolates this functional role: a single-batch loss probe with no access to direction labels accepts $\sim 70\%$ of cache injections built from tail/floor directions (loss decreases on a separate local batch) and rejects 100% of cache injections built from head/middle directions (loss increases). This direction-blind asymmetry directly shows that restoring a tail/floor direction reduces loss for its owner client, while restoring a head/middle direction does not.

Together, these observations show that factor-wise aggregation systematically destroys the tail/floor directions encoding client-specific adaptation. The persistent zigzag in Figure 2 further shows that this knowledge loss is not a one-time event: the same directions are re-learned every round only for the next aggregation to erase them again, creating an avoidable recovery cost at every participation.

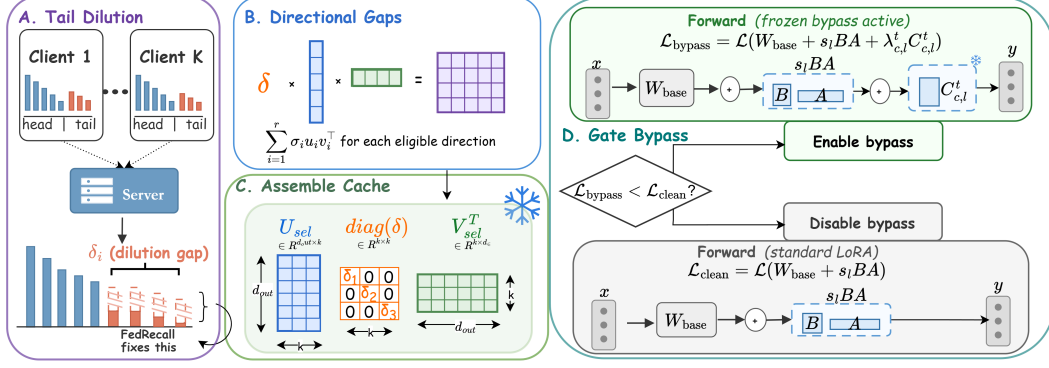


Figure 3: **Overview of FedReCall.** Factor-wise aggregation creates dilution gaps in weak client-specific directions. After aggregation, each client estimates the missing directional strength, assembles the reliable gaps into a private frozen cache, and reuses the cache as a low-rank bypass in later local training. A one-batch gate enables the bypass only when it lowers the client’s local loss; otherwise the client falls back to standard LoRA.

168 Rather than redesigning the server aggregator—which would risk disrupting the cross-client consensus
 169 that preserves head directions—we keep these diluted directions as private memory on the clients
 170 that originated them.

171 3.3 FedReCall: client-private directional recall

172 Guided by the dilution analysis above, FedReCall protects diluted tail/floor directions as client-private
 173 memory rather than as shared parameters. It is a personalization plugin on top of standard factor-wise
 174 FedIT: clients still upload only clean LoRA factors, and the server aggregation rule is unchanged.
 175 After each aggregation a participating client builds a correction cache (**cache construction**); when
 176 next selected, it loads the cache as a frozen bypass during local adaptation (**bypass injection**), gated
 177 against stale caches by a one-batch loss probe (**quality gate**).

178 **Cache construction.** After client c completes local training in round t and receives the new post-
 179 aggregation global factors $(A_l^{t+1, \text{glob}}, B_l^{t+1, \text{glob}})$, it builds the correction cache in three steps. First,
 180 the client decomposes its *effective update* via SVD to obtain the singular triplets $\{(\sigma_i, u_i, v_i)\}_{i=0}^{r-1}$.
 181 The effective update captures both the trained LoRA factors and any active bypass from the current
 182 round,

$$\widetilde{\Delta W}_{c,l}^t = s_l B_{c,l}^t A_{c,l}^t + m_c^t \lambda_{c,l}^{\text{prev}} C_{c,l}^{\text{prev}}, \quad (7)$$

183 where $m_c^t \in \{0, 1\}$ indicates whether the bypass was enabled in round t and $C_{c,l}^{\text{prev}}$ is the cache from
 184 the client’s most recent prior participation; for a first-time client, $m_c^t = 0$ and the effective update
 185 reduces to the trained LoRA factors alone. Second, it computes the global retention α_i and dilution
 186 gap δ_i (Eqs. 5–6) for each direction, using the Frobenius projection defined in Section 3.2; the value
 187 δ_i estimates how much client signal is missing from the current global model along direction ϕ_i and
 188 becomes the amplitude stored in the cache. Third, motivated by the finding that tail/floor directions
 189 are both more strongly diluted and more client-specific (Section 3.2), it selects the eligible tail/floor
 190 set:

$$I_{c,l}^{t, \text{elig}} = \{i : \delta_i > 0, \sigma_i \geq \tau_l, i \in \text{tail} \cup \text{floor}\}, \quad (8)$$

191 where $\tau_l = \epsilon_{\text{rank}} \sigma_{l, \text{max}}$ with $\epsilon_{\text{rank}} = 10^{-4}$ excludes near-zero directions, and the index i is restricted
 192 to the tail and floor bands defined in Section 3.1. For LoRA rank $r = 32$, this restriction picks up
 193 indices $i \geq 20$, yielding up to 12 candidate directions per layer. The private correction cache for
 194 layer l is

$$C_{c,l}^t = \sum_{i \in I_{c,l}^{t, \text{elig}}} \delta_i u_i v_i^\top. \quad (9)$$

195 The cache stores at most 12 triplets per layer (negligible vs. the LoRA factors), and SVD is computed
 196 via QR-compressed low-rank algebra (Appendix E).

197 **Bypass injection and calibration.** When
 198 client c is next selected, it loads the cached cor-
 199 rection and injects it as a frozen additive bypass
 200 in each layer’s forward pass:

$$y = W_{\text{base}}x + s_l B_l A_l x + \lambda_{c,l} C_{c,l} x. \quad (10)$$

201 During local training, only A_l and B_l are op-
 202 timized; $C_{c,l}$ is frozen and excluded from the
 203 optimizer, and the gradient pathways to A_l and
 204 B_l are unchanged (Appendix F). The bypass
 205 exploits the low-rank structure of $C_{c,l}$:

$$C_{c,l}x = U_{\text{sel}}[\text{diag}(\delta)(V_{\text{sel}}^\top x)], \quad (11)$$

206 where U_{sel} and V_{sel} stack the selected left and
 207 right singular vectors. The bypass strength is
 208 calibrated relative to the clean LoRA update
 209 norm via a global hyperparameter η :

$$\lambda_{c,l} = \eta \cdot \frac{\|s_l B_l A_l\|_F}{\|C_{c,l}\|_F + \epsilon}. \quad (12)$$

210 Because the selected direction atoms are
 211 Frobenius-orthonormal, $\|C_{c,l}\|_F = \|\delta\|_2$, which can be computed without materializing
 212 the full correction matrix. This calibration ensures that the bypass-to-LoRA norm ratio
 213 $\|\lambda_{c,l} C_{c,l}\|_F / \|s_l B_l A_l\|_F \approx \eta$ holds consistently across layers and clients regardless of their ab-
 214 solute scales (derivation in Appendix G). A single global η thus suffices without per-layer tuning ($\eta =$
 215 0 recovers standard federated LoRA); η sensitivity is studied in Section 4.3.

216 **One-batch quality gate.** A client’s cached correction may become stale if the global model has
 217 evolved substantially since the cache was built. Before enabling the bypass, FedReCall evaluates a
 218 single local mini-batch $\mathcal{B}_{\text{gate}}$ under both the model without bypass and the bypassed model:

$$m_c^t = \mathbf{1}[\mathcal{L}(\theta^{\text{LoRA}}, \Lambda_c \mathbf{C}_c; \mathcal{B}_{\text{gate}}) < \mathcal{L}(\theta^{\text{LoRA}}, \mathbf{0}; \mathcal{B}_{\text{gate}})], \quad (13)$$

219 where $\theta^{\text{LoRA}} = \{(A_l, B_l)\}_l$ denotes the shared LoRA factors, $\Lambda_c \mathbf{C}_c = \{\lambda_{c,l} C_{c,l}\}_l$ denotes the
 220 per-layer frozen bypass, and $\mathbf{0}$ indicates the bypass is disabled. If the gate rejects the cache, the client
 221 trains with standard LoRA for that round. The gate’s overhead is a single forward pass per model
 222 variant.

223 The full per-round procedure is summarized in Algorithm 1; beyond standard federated LoRA,
 224 FedReCall adds only the client-side cache construction (Phase 3) and bypass injection (Phase 1), with
 225 the server aggregation unchanged and a single hyperparameter η .

226 4 Experiments

227 4.1 Experimental setup

228 Following Bai et al. [2], we construct a task-heterogeneous federated setting from the Super-Natural
 229 Instructions (SuperNI) dataset [33]. The full pool contains 511 NLP tasks from 76 task types; each
 230 client holds a single task. We randomly draw 50 clients (seed 42) as the training federation; details of
 231 the client selection are provided in Appendix H. This federation is intentionally heterogeneous: each
 232 client corresponds to a distinct SuperNI task, the selected 50-client cohort spans 23 task types, and
 233 client data volumes are highly uneven. The base model is DataJuicer-LLaMA-1.3B [32], fine-tuned
 234 with LoRA [12] ($r=32$); all methods share the same rank for fair comparison. Unless otherwise
 235 stated, all reported numbers are averaged over three training seeds. We compare against two families
 236 of baselines. (i) *Aggregation-side methods*—FedIT [42], FLoRA [34], FFA-LoRA [30], and FedSA-
 237 LoRA [4]—differ in how they aggregate LoRA factors; FedReCall is layered on top of each as a
 238 plug-in (Table 2). (ii) *Dual-LoRA personalization methods*—FedDPA-T [38] and FDLORA [26]—
 239 each maintain a global and a local adapter; they serve as comparison points for the personalization
 240 capability of FedReCall.

Algorithm 1 FedReCall: client c , round t

Require: $\eta > 0$
 1: **Input:** $\{A_l^t, B_l^t\}_l$, $\mathcal{D}_c$; optional C_c^{prev}
 2: // Phase 1: Bypass injection
 3: **if** C_c^{prev} exists **then**
 4: $\lambda_{c,l} \leftarrow \eta \|s_l B_l^t A_l^t\|_F / (\|C_{c,l}^{\text{prev}}\|_F + \epsilon)$
 5: $m_c^t \leftarrow \mathbf{1}[\mathcal{L}(\theta, \lambda C; \mathcal{B}) < \mathcal{L}(\theta, \mathbf{0}; \mathcal{B})]$
 6: **if** $m_c^t = 1$ **then**
 7: $y \leftarrow \lambda_{c,l} C_{c,l}^{\text{prev}} x$ (frozen)
 8: **end if**
 9: **else**
 10: $m_c^t \leftarrow 0$ (bootstrap)
 11: **end if**
 12: // Phase 2: Local training
 13: $\{(A_{c,l}^t, B_{c,l}^t)\}_l \leftarrow \text{SGD}(\{A_l, B_l\}_l; \mathcal{D}_c)$
 14: Upload $\{(A_{c,l}^t, B_{c,l}^t)\}_l$
 15: // Phase 3: Cache construction
 16: Receive $\{A_l^{t+1}, B_l^{t+1}\}_l$; compute $\widetilde{\Delta W}_{c,l}^t$
 17: $\alpha_i \leftarrow \langle \Delta W_l^{t+1, \text{glob}}, \phi_i \rangle_F$; $\delta_i \leftarrow \max(0, \sigma_i - \alpha_i^+)$
 18: $I_{c,l}^t \leftarrow \{i : \delta_i > 0, i \in \text{tail} \cup \text{floor}\}$
 19: $C_{c,l}^t \leftarrow \sum_{i \in I_{c,l}^t} \delta_i u_i v_i^\top$; store locally

Table 2: Main performance on the task-heterogeneous federation.

Method	Personalized client model			Global model		
	Avg Pers. Loss ↓	Worst Loss ↓	Pers. ROUGE-L ↑	Avg Loss ↓	ROUGE-L ↑	Pers. Gap
FedIT	1.225	2.201	70.63	2.252	61.04	1.027
+ FedReCall	1.035	2.130	74.04	2.029	66.41	0.922
FLoRA	1.273	1.460	70.85	2.397	67.50	1.124
+ FedReCall	1.146	1.406	82.48	1.807	69.44	0.661
FFA-LoRA	1.345	1.960	56.72	1.985	71.28	0.640
+ FedReCall	1.089	1.554	84.47	1.244	80.58	0.155
FedSA-LoRA	1.291	2.168	69.70	2.045	62.33	0.754
+ FedReCall	1.236	2.164	75.27	1.414	77.16	0.178
<i>Dual-LoRA personalized baselines</i>						
FedDPA-T (1×)	1.809	2.035	57.51	2.111	57.93	0.302
FedDPA-T (2×)	1.177	2.001	73.25	2.390	51.64	1.213
FDLoRA (1×)	1.326	2.028	70.45	2.414	64.56	1.088

241 *Personalized* metrics (Avg Pers. Loss, Avg Pers. ROUGE-L) measure each client’s post-local-
 242 adaptation performance. *Global* metrics (Avg Loss, Avg ROUGE-L) assess the global model before
 243 any local adaptation, tracking how the plug-in influences the shared model.

244 4.2 Main results: FedReCall as a plug-in

245 Table 2 reports all baselines on the 50-client SuperNI-derived federation, with FedReCall applied as
 246 an additive plug-in on top of each aggregator. We report both client-side metrics after local adaptation
 247 and global-model metrics before any local adaptation. We also include the personalization gap, i.e.,
 248 global-model loss minus average personalized loss. FedReCall uses the same candidate grid for its
 249 single global bypass-strength parameter η across all plug-in settings; the selected values for Table 2
 250 are $\eta=10$ for FedIT, $\eta=2$ for FLoRA, $\eta=20$ for FFA-LoRA, and $\eta=30$ for FedSA-LoRA. The full
 251 single-seed sweep is reported in Appendix I; no per-client or per-layer η tuning is used.

252 FedReCall consistently improves personalized adaptation as a plug-in.

253 Table 2 shows that
 254 adding FedReCall improves average personal-
 255 ized loss and personalized ROUGE-L for every
 256 aggregation-side baseline. The improvement is not
 257 limited to the standard FedIT setting: it also ap-
 258 pears when the base method changes which LoRA
 259 factors are shared, fixed, or aggregated. FedSA-
 260 LoRA, for example, keeps B local, which can
 261 reduce the exposure of B -side client-specific in-
 262 formation to direct averaging. However, this is
 263 a factor-level design choice, whereas FedReCall
 264 identifies and recalls diluted directions in the ef-
 265 fective update $\Delta W = BA$. The strongest gains
 266 occur for FFA-LoRA and FLoRA, while FedSA-
 267 LoRA shows a smaller but still positive personal-
 268 ized improvement. This pattern is consistent with
 269 the view that FedReCall is most useful when the base aggregator leaves an observable directional gap
 270 to repair; Section 4.3 examines this interpretation through the η sensitivity and fragility diagnostics.
 271 A same-round paired subgroup diagnostic in Appendix D further indicates that the gains are not
 272 concentrated in a few clients, but remain visible across data-volume and baseline-difficulty groups.

273 Dual-LoRA personalized baselines split the local training budget between a global and a local
 274 adapter; under matched compute, both undertrain (FedDPA-T 1× loss 1.81, FDLoRA 1.33). Even
 275 at 2× compute, FedDPA-T (loss 1.18) does not catch up to FedReCall, while adding zero trainable
 276 parameters and preserving the single-LoRA protocol.

Table 3: Zero-shot performance on unseen tasks.

Method	Avg. ROUGE-L ↑	Avg. loss ↓	Worst loss ↓
FedIT	50.27	2.26	2.83
+ FedReCall	51.14	2.19	2.58
FLoRA	51.53	2.22	2.72
+ FedReCall	52.51	2.26	2.66
FFA-LoRA	54.75	1.99	2.47
+ FedReCall	55.62	1.95	2.43
FedDPA-T (1×)	52.69	2.09	2.50
FedDPA-T (2×)	51.86	2.25	2.73
FDLoRA (1×)	50.41	2.42	3.08

Main: Fixed $\eta = 10$ across all aggregators.

FedReCall improves global model quality alongside personalization. FedReCall also improves global model quality across all four aggregators, despite its cache never being uploaded—the server-side aggregation is mechanically identical to the base method. The global gain therefore cannot come from the server directly receiving private cached directions. We interpret it as a change in the burden placed on the next round of uploaded clean LoRA factors: once recurring private corrections are recalled locally, the trainable factors can carry more shared adaptation signal while the non-shared correction remains local. The effect is most pronounced on FFA-LoRA, where global ROUGE-L rises by +9.3 points and global loss drops by 37%. This dual improvement—personalized *and* global—makes FedReCall the only method in Table 2 that reaches the upper-right corner of strong performance on both axes. FedReCall introduces additional client-side computation for cache construction and gate evaluation; detailed wall-clock measurements are reported in Appendix J.

FedReCall preserves global-model generalization. Table 3 evaluates the final global model on unseen task types without local adaptation or private recall. The results show that the improved global metrics in Table 2 are not an artifact of evaluating with private memory enabled: the final global model remains competitive on unseen task types while the cache stays strictly local.

Table 4: FedReCall component ablation.

Bypass	Gate	Avg. Pers. loss ↓	Global loss ↓	Δ vs. FedIT
—	—	1.225	2.253	—
✓	—	1.115	2.329	−9.0%
✓	✓	1.035	2.029	−15.5%

4.3 Ablation and analysis

We next test whether FedReCall’s design choices are necessary for repairing directional dilution rather than merely adding extra local capacity. The ablations follow the mechanism in Section 3.2: a useful correction should restore diluted directions locally, reject stale private signal, and operate on the effective update ΔW where dilution is defined. A compact operand ablation in Appendix A further verifies that the cache should be built from $\Delta W = BA$ rather than from an individual LoRA factor.

Component ablation and gate validation. The frozen bypass and the quality gate play different roles. The bypass supplies the missing directions during local adaptation, which accounts for most of the personalized improvement. The gate turns this memory into a validated correction: a cached direction is used only when it improves the current client’s mini-batch loss. This matters because early caches can be stale or over-amplified; without the gate, recall remains helpful for personalization but can degrade the shared model. Table 6 in Appendix B further validates this behaviour using candidate pools that the gate does not observe. Low-strength candidates are often accepted and improve bypassed loss, whereas high/mid-strength candidates are rejected because they sharply increase loss, confirming that the gate performs functional validation rather than merely following a spectral label.

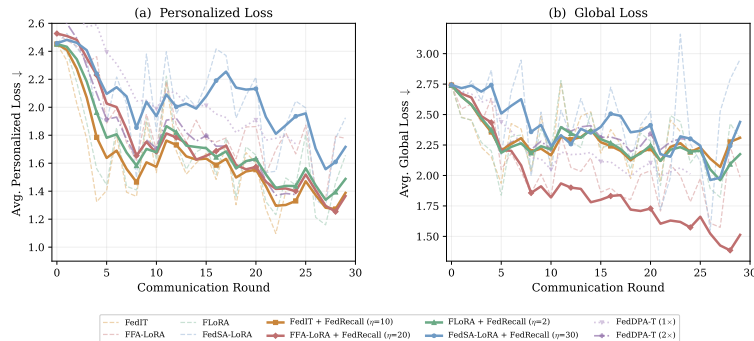


Figure 4: **Training dynamics with and without FedReCall.** Dashed curves are base aggregators; solid curves add FedReCall, with EMA smoothing used only for visualization.

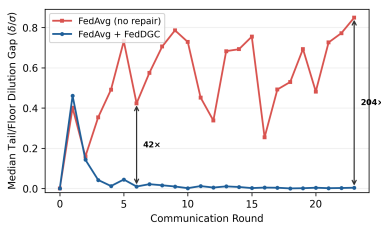


Figure 5: **Fragility repair diagnostic.**

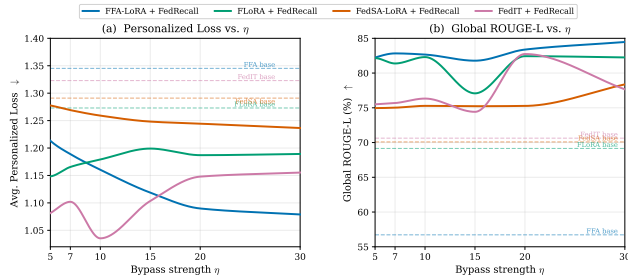


Figure 6: **Bypass strength sensitivity.** The useful η range depends on the base aggregator.

Training dynamics and fragility repair. Figure 4 traces the per-round personalized and global loss across all compared methods. Baselines without FedReCall show round-to-round oscillations consistent with the recovery–disruption cycle in Section 3.2; FedReCall dampens these oscillations and converges to lower loss. Figure 5 tracks the same relative dilution gap as Figure 2(a) across rounds with FedReCall enabled. The gap moves toward zero, indicating that FedReCall reduces the same directional dilution measured in the motivation rather than merely adding black-box local capacity.

Sensitivity and robustness. Figure 6 sweeps the bypass strength η . FedReCall remains effective over a non-trivial range of η , with the preferred value tracking how much directional dilution the base aggregator leaves: aggregators with stronger dilution benefit from stronger recall, while aggregators that already preserve more private signal peak at moderate η . This indicates η is a coarse global compensation knob rather than a narrow optimum. Appendix C reports additional diagnostics over LoRA rank and client participation rate. The rank sweep checks whether recall becomes more useful as the adapter exposes more weak local directions, while the participation-rate sweep checks whether the effect persists when more clients are selected per communication round. Across ranks 16/32/64, FedReCall matches or improves FedIT, with the largest gain at higher rank; across participation rates 10–30%, it preserves a positive ROUGE-L gap while the loss gap narrows as the base FedIT run strengthens. These diagnostics support the same mechanism-level interpretation without overloading the main text with auxiliary tables.

5 Conclusion

Taken together, the analysis and experiments show that personalization loss in federated LoRA is tied to a concrete aggregation effect: under task heterogeneity, factor-wise aggregation repeatedly weakens locally useful directions that are weakly shared across clients. FedReCall turns this mechanism-level observation into a client-side repair. It estimates observable dilution gaps after aggregation, stores reliable missing directions in a private frozen cache, and recalls them through a gated low-rank bypass during later local adaptation. Because the cache is neither uploaded nor optimized, FedReCall leaves the server aggregation rule, communication payload, and trainable-parameter set unchanged. Across representative FedLoRA aggregators, the results show that selectively restoring diluted client-specific directions improves personalized adaptation and also strengthens the global model in the main experiments, while remaining competitive on unseen zero-shot evaluation.

Limitations. We have not yet evaluated FedReCall on a broader range of datasets and downstream tasks. This leaves open how well the same directional-recall principle generalizes beyond the task-heterogeneous LoRA instruction-tuning setting studied here. Finally, although the cache is never uploaded, it remains client-specific state. Studying its interaction with formal privacy guarantees, secure local storage, other adapter families, and larger model scales is an important direction for future work.

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A Operand ablation

Table 5: Operand ablation on FFA-LoRA. Applying FedReCall to the effective update $\Delta W = BA$ yields clear average-loss and ROUGE-L gains, whereas a naive B -only variant provides no benefit.

Method	Weighted Loss ↓	ROUGE-L ↑
FFA-LoRA	1.393	0.684
+ FedReCall (B -only, naive)	1.405	67.55
+ FedReCall ($\Delta W = BA$, principled)	1.083	77.23

484 B Gate behaviour details

Table 6: Gate behaviour under direction-strength candidate pools.

Candidate range	Gate pass rate	No-bypass loss	Bypassed loss
Low-strength range ($q_i \geq 0.6$)	0.78 ± 0.19	2.00 ± 0.04	1.95 ± 0.04
Low-strength range, uniform amplitude	1.00 ± 0.00	2.07 ± 0.10	1.98 ± 0.07
High/mid-strength range ($q_i < 0.6$)	0.00 ± 0.00	1.96 ± 0.11	11.51 ± 0.33

485 C Robustness diagnostics

486 Table 7 reports a LoRA-rank diagnostic using the same federation and client-selection seed. The
 487 comparison asks whether directional recall is tied to the default rank or becomes more useful as the
 488 adapter exposes more weak local directions. Table 8 complements the default 10% participation
 489 setting by increasing the number of clients selected in each round. This diagnostic keeps the same
 490 50-client federation and client-selection seed, runs for 15 communication rounds, and compares all
 491 runs at round 10.

Table 7: LoRA-rank robustness diagnostic.

Rank	Method	Avg Pers. Loss ↓	Pers. ROUGE-L ↑
16	FedIT	1.146	83.96
	+ FedReCall	1.144	83.55
32	FedIT	1.225	70.63
	+ FedReCall	1.035	74.04
64	FedIT	1.039	76.18
	+ FedReCall	0.960	77.88

Table 8: Participation-rate robustness diagnostic.

Part.	Method	Avg Pers. Loss ↓	Pers. ROUGE-L ↑
10%	FedIT	1.540	57.43
	+ FedReCall	1.0354	74.04
20%	FedIT	1.39	68.61
	+ FedReCall	1.3906	70.28
30%	FedIT	1.401	66.32
	+ FedReCall	1.448	67.60

492 D Same-round subgroup diagnostic

493 To check whether the main gains are driven by a few favorable clients, we run a same-round paired
 494 diagnostic on the selected clients used by each base method and its FedReCall plug-in. For each
 495 paired client, we compute ΔLoss as base loss minus FedReCall loss and $\Delta\text{ROUGE-L}$ as FedReCall
 496 ROUGE-L minus base ROUGE-L. We then group the paired clients by data volume and by baseline
 497 difficulty. This analysis is descriptive because each round selects only five clients, but it verifies that
 498 the improvement is not solely an artifact of comparing different client sets.

Table 9: Same-round paired subgroup diagnostic. Positive ΔLoss means FedReCall lowers client loss. The last two columns count how many data-volume and baseline-difficulty groups have positive mean ΔLoss .

Base method	Paired clients	$\Delta\text{Loss} \uparrow$	$\Delta\text{ROUGE-L} \uparrow$	Data-volume groups	Difficulty groups
FedIT	5	0.2881	5.70	3/3	3/3
FLoRA	5	0.0127	0.45	2/3	2/3
FFA-LoRA	5	0.2554	4.93	3/3	3/3
FedSA-LoRA	5	0.1016	3.09	2/3	2/3

E QR-compressed SVD for cache construction

The cache construction requires the SVD of the effective update $\widetilde{\Delta W}_{c,l}^t \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ (Eq. 7), which can be prohibitively expensive if materialized as a dense matrix. We exploit the low-rank factored form to avoid this.

When no bypass is active ($m_c^t = 0$), the effective update is $s_l B_{c,l}^t A_{c,l}^t$, which already has rank at most r . When the bypass is active, we have

$$\widetilde{\Delta W}_{c,l}^t = \underbrace{[s_l B_{c,l}^t \mid \lambda_{c,l}^{\text{prev}} U_{\text{sel}} \text{diag}(\delta^{\text{prev}})]}_{L \in \mathbb{R}^{d_{\text{out}} \times (r+k)}} \underbrace{\begin{bmatrix} A_{c,l}^t \\ V_{\text{sel}}^{\text{prev}^\top} \end{bmatrix}}_{R \in \mathbb{R}^{(r+k) \times d_{\text{in}}}},$$

where $k = |I_{c,l}^{\text{prev}}|$ is the number of cached tail/floor directions (at most $|\text{tail}| + |\text{floor}|$). The product LR has rank at most $r + k$, and its SVD can be obtained without forming the $d_{\text{out}} \times d_{\text{in}}$ matrix:

1. Compute a thin QR factorization $L = Q_L R_L$, where $Q_L \in \mathbb{R}^{d_{\text{out}} \times (r+k)}$ has orthonormal columns and $R_L \in \mathbb{R}^{(r+k) \times (r+k)}$.
2. Form the small matrix $M = R_L R \in \mathbb{R}^{(r+k) \times d_{\text{in}}}$.
3. Compute the SVD $M = U_M \Sigma V^\top$.
4. The left singular vectors of $\widetilde{\Delta W}_{c,l}^t$ are $Q_L U_M$; the singular values and right singular vectors are Σ and V .

All operations involve matrices of size at most $(r + k) \times \max(d_{\text{out}}, d_{\text{in}})$, giving a total cost of $O((r + k)^2(d_{\text{out}} + d_{\text{in}}))$ per layer, compared to $O(d_{\text{out}} d_{\text{in}} \min(d_{\text{out}}, d_{\text{in}}))$ for a dense SVD.

F Gradient flow through the frozen bypass

A natural concern is whether the frozen bypass distorts the gradient signal received by the trainable LoRA factors. We show that it does not block or redirect any gradient pathway.

The bypassed forward pass for layer l is (Eq. 10)

$$y = W_{\text{base}} x + s_l B_l A_l x + \lambda_{c,l} C_{c,l} x.$$

Because $C_{c,l}$ is a frozen constant, the partial derivatives of y with respect to the trainable factors are

$$\frac{\partial y}{\partial B_l} = s_l (A_l x) \otimes I, \quad \frac{\partial y}{\partial A_l} = s_l B_l^\top \otimes x^\top,$$

which are identical to the standard LoRA case without bypass. The bypass term $\lambda_{c,l} C_{c,l} x$ affects the loss $\mathcal{L}$ by changing the output y , and therefore changes the upstream gradient $\partial \mathcal{L} / \partial y$. However, the Jacobians $\partial y / \partial B_l$ and $\partial y / \partial A_l$ are bypass-independent.

In effect, the frozen bypass shifts the output baseline from which LoRA operates: during local adaptation, the LoRA factors need not spend capacity re-learning the directions already restored by the cache, and can focus on further task-specific adaptation. No stop-gradient is applied to the shared input x , so gradient information flows freely to all layers below.

G Bypass strength calibration derivation

We show that the per-layer scaling $\lambda_{c,l}$ in Eq. 12 achieves the target bypass-to-LoRA norm ratio η up to negligible ϵ -stabilizer terms.

Setup. The cache admits the factorization $C_{c,l} = U_{\text{sel}} \text{diag}(\boldsymbol{\delta}) V_{\text{sel}}^\top$, where the columns of U_{sel} and V_{sel} are subsets of orthonormal singular vectors. Because distinct direction atoms $\phi_i = u_i v_i^\top$ are Frobenius-orthogonal,

$$\|C_{c,l}\|_F^2 = \sum_{i \in I_{c,l}^{\text{elig}}} \delta_i^2 = \|\boldsymbol{\delta}\|_2^2.$$

Derivation. Substituting Eq. 12 into the bypass norm:

$$\|\lambda_{c,l} C_{c,l}\|_F = \lambda_{c,l} \|C_{c,l}\|_F = \eta \cdot \frac{\|s_l B_l A_l\|_F}{\|C_{c,l}\|_F + \epsilon} \cdot \|C_{c,l}\|_F.$$

Dividing by $\|s_l B_l A_l\|_F$:

$$\frac{\|\lambda_{c,l} C_{c,l}\|_F}{\|s_l B_l A_l\|_F} = \eta \cdot \frac{\|C_{c,l}\|_F}{\|C_{c,l}\|_F + \epsilon}.$$

When $\|C_{c,l}\|_F \gg \epsilon$ (which holds for any non-trivial cache), the ratio approaches η . The ϵ term (set to 10^{-8} in practice) serves only as a numerical guard against division by zero and does not affect the calibration for any layer with a non-empty cache.

H Data preparation and client selection

Client pool. Following Bai et al. [2], we use Super-Natural Instructions [33] to construct a task-heterogeneous federation of 511 clients covering 76 NLP task types. Each client holds a single task, and within-task training data is subsampled to 10% following the original FlexLoRA protocol.

50-client subset. From the 511-client pool we draw 50 clients uniformly at random (`random.sample(range(511), 50, seed=42)`) and re-index them as clients $0, \dots, 49$. A deterministic mapping from new to original client IDs is stored for reproducibility. The selected 50 clients span 23 NLP task types; per-client training-set sizes range from 16 to 520 samples (median 127, mean 232).

Per-round participation. In each communication round, 5 clients (10%) are sampled without replacement. The client-selection seed is fixed to 42 and shared across all training seeds $\{42, 43, 44\}$, so that every run uses an identical participation schedule. This design isolates the reported variance to training randomness alone, excluding participation-schedule noise.

I Bypass strength selection

Table 10 reports a single-seed diagnostic sweep over the bypass-strength parameter η (seed 42, $q=0.6$, 30 rounds). The purpose is to characterize how different base aggregators respond to stronger or weaker private recall, not to claim a monotonic trend or introduce per-layer tuning. Since η is applied through the norm calibration in Eq. 12, it acts as one coarse global compensation factor across layers and clients.

Table 10: Single-seed bypass-strength sensitivity.

η	2	3.5	5	7	10	15	20	30
FFA-LoRA	1.282	1.248	1.213	1.189	1.160	1.119	1.090	1.081
FLoRA	1.147	1.176	1.168	1.186	1.179	1.199	1.167	1.1892
FedSA-LoRA	1.285	1.282	1.278	1.269	1.259	1.248	1.244	1.237
FedIT	1.2823	1.2484	1.081	1.102	1.035	1.104	1.148	1.155

J Computational infrastructure

All experiments are conducted on a single NVIDIA GeForce RTX 5090 GPU. The codebase is built on PyTorch 2.7.0 (CUDA 12.8).

Client-side compute and storage. FedReCall does not change the communication payload or the trainable-parameter set, but it does add client-side computation and private state. The extra computation comes from QR-compressed SVD cache construction and a one-batch quality gate. The gate reuses a local mini-batch already available on the client; it does not require uploading data or storing a separate gate dataset. The private cache stores selected triplets (δ_i, u_i, v_i) and remains local to the client. Table 11 reports the average per-round wall-clock time measured over the stable training phase (excluding initial warm-up rounds).

Table 11: Client-side wall-clock overhead of FedReCall. The increment is measured relative to each corresponding base method, not relative to FedIT.

Base method	Base (min)	+ FedReCall (min)	Increment
FedIT	3.4	4.7	+1.3 min / +38%
FFA-LoRA	4.1	4.9	+0.8 min / +20%
FLoRA	5.5	6.5	+1.0 min / +18%
FedSA-LoRA	4.4	4.7	+0.3 min / +7%
<i>Dual-LoRA personalized baselines, no FedReCall</i>			
FedDPA-T (1×)		5.1	–
FedDPA-T (2×)		6.3	–
FDLORA		4.8	–

NeurIPS Paper Checklist

1. Claims

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